Photon Shower Shape Corrections Using Cell-Level Energy Reweighting with ATLAS Run 3 Data

Marta Fernandes da Silva

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Abstract

Electromagnetic shower shapes (R_η, R_ϕ, $w_{\eta 2}$) are critical inputs to photon identification in ATLAS. Residual data–MC mismodelling of the calorimeter cell energy deposits degrades the accuracy of these discriminating variables. This note presents a cell-level energy reweighting approach applied to Layer 2 of the electromagnetic calorimeter using $Z \to \ell\ell\gamma$ final-state-radiation events from ATLAS Run 3 data (2024) and MC23e simulation.

Two correction methods are studied. Method 1 (M1, flat shift) adjusts each cell energy by the mean data–MC fraction difference. Method 2 (M2, shift+stretch) additionally matches the variance by applying a linear transformation derived from Francisco Almeida's `photoncellbasedrw` framework. Both methods are evaluated across 13 non-crack $|\eta|$ bins, the combined $Z \to \ell\ell\gamma$ channel, three photon conversion scenarios (unconverted, converted, inclusive), and four pipeline variants covering $|\eta|$-only and two-dimensional ($|\eta|$, p_{T}) correction binning with loose and tight isolation.

An initial implementation of M2 suffered from catastrophic instabilities in specific $|\eta|$ bins, where divergent variance ratios in peripheral cells produced χ^2/n_{df} values up to 90× worse than uncorrected MC. Three targeted fixes — histogram-based σ estimation via `GetRMS()`, cell-dependent histogram ranges acting as natural regularisation, and removal of a minimum-fraction fallback — completely eliminated these instabilities. The stabilised M2 now outperforms M1 for all three shower shape variables, with average χ^2/n_{df} improvements of 17% (R_η), 45% (R_ϕ), and 29% ($w_{\eta 2}$) relative to M1. M2 nearly matches the empirical ATLAS fudge factors, achieving identical χ^2/n_{df} for R_η and $w_{\eta 2}$ and within 20% for R_ϕ.

A cross-source comparison with Francisco Almeida's `photoncellbasedrw` Run 3 output confirms equivalent correction quality, with small residual χ^2/n_{df} differences attributable to different histogram axis ranges used in the two implementations. M2 with cell-dependent histogram ranges is the recommended cell-level correction method. The `eta_pt_loose` variant (14 $|\eta|$ × 6 p_{T} bins, loose isolation) achieves the best correction quality with $\langle \chi^2/n_{\mathrm{df}} \rangle = 0.0003$ for all three shower shape variables in the inclusive scenario.

Contents

1 Introduction **2**

2 Data and MC Samples **2**

3 Event Selection **2**

3.1 NTupleMaker-level preselection 2

3.2 Offline selection 4

3.3 Conversion scenarios 4

4 Shower Shape Variables **4**

5 Correction Methods **4**
5.1 Method 1: flat shift . . . 6
5.2 Method 2: shift+stretch (Francisco's method) . . . 6
5.2.1 Variance estimation: histogram-based σ via `GetRMS()` . . . 6
5.2.2 Cell-dependent histogram ranges . . . 7
5.2.3 Removal of the minimum-fraction fallback . . . 8
5.3 Pipeline variants: binning and isolation . . . 9

6 Results: Unconverted Photons **9**
6.1 Shower shape comparisons . . . 10
6.2 Chi-squared comparison . . . 10
6.3 Average performance . . . 10

7 Conversion Scenarios **13**
7.1 Converted photons . . . 13
7.2 Inclusive . . . 13
7.3 Comparison across scenarios . . . 13

8 Cell Energy Profile Maps **15**
8.1 Before and after correction . . . 15
8.2 M2 correction vectors . . . 15

9 Discussion **15**
9.1 M1 performance . . . 17
9.2 M2 stabilisation: from catastrophic failures to best-in-class performance . . . 18
9.2.1 Previous M2 behaviour (initial implementation) . . . 18
9.2.2 The three-pronged fix . . . 18
9.2.3 Stabilised M2 performance . . . 19
9.3 Cross-source comparison with Francisco's `photoncellbasedrw` . . . 20
9.3.1 Methodology . . . 20
9.3.2 Histogram axis range effect . . . 20
9.4 Comparison with fudge factors . . . 21
9.5 Wide vs. cell-dependent histogram ranges . . . 21
9.6 Problematic $|\eta|$ regions . . . 22
9.7 Isolation requirement . . . 22
9.8 Cross-variant comparison . . . 22

10 Summary **23**

A Per-Eta Chi-Squared Tables **24**

B Cross-Source Comparison Per-Eta Tables **27**

1 Introduction

Photon identification in ATLAS relies on shower shape variables computed from the energy deposits in the electromagnetic calorimeter. In particular, R_η, R_ϕ, and $w_{\eta 2}$ are built from the Layer 2 cell energies in a 7×11 ($\eta \times \phi$) cluster and are among the most discriminating variables for separating prompt photons from hadronic fakes.

Residual mismodelling of these variables in Monte Carlo simulation is addressed by the ATLAS "fudge factor" framework, which applies empirical corrections to the branch-level shower shape values. An alternative approach, pioneered by Francisco Almeida in Run 2 [1], operates at the cell-energy level: the individual cell energies are corrected before recomputing the shower shapes, thereby preserving the physical correlations between cells.

This note documents the Run 3 implementation of the cell-level reweighting approach. Two methods are studied:

- **Method 1 (M1)**: a flat shift per cell, matching the population mean of the cell energy fraction.
- **Method 2 (M2)**: a shift+stretch transformation, matching both the mean and the variance, following Francisco Almeida's `photoncellbasedrw` method [1].

The analysis uses $Z \to \ell\ell\gamma$ events selected from ATLAS 2024 data and MC23e simulation, processed through the NTupleMaker framework.

An earlier version of this pipeline found M2 to be unstable, with catastrophic failures in certain $|\eta|$ bins traced to divergent variance ratios in peripheral cells. This note describes the three code improvements that eliminated those instabilities and presents the resulting performance, where M2 is now the recommended cell-level correction method.

2 Data and MC Samples

The analysis uses ATLAS data collected in 2024 and the corresponding MC23e simulation campaign. Two channels are considered:

- $Z \to ee\gamma$: data from DAOD_EGAM3 (`egam3.root`), MC sample DSID 700770.
- $Z \to \mu\mu\gamma$: data from DAOD_EGAM4 (`egam4.root`), MC sample DSID 700771.

The combined channel ($Z \to \ell\ell\gamma$) merges both data and MC samples. Ntuples are produced by the NTupleMaker package running in Athena 25.0.40.

MC events are weighted by $w = w_{\mathrm{MC}} \times w_\mu \times \sigma \times \mathrm{SF}_{\mathrm{trig}} \times \mathrm{SF}_\gamma \times \mathrm{SF}_\ell$, where w_{MC} is the generator weight, w_μ the pile-up reweighting factor, σ the cross-section weight, and $\mathrm{SF}_{\mathrm{trig}}$, SF_γ, SF_ℓ are the trigger, photon, and lepton scale factors respectively.

3 Event Selection

Events are selected in two stages: a preselection applied at the NTupleMaker level during ntuple production, and an offline selection applied by the analysis pipeline.

3.1 NTupleMaker-level preselection

The NTupleMaker applies Good Runs List (GRL) filtering, trigger matching, and basic lepton quality requirements. Photon candidates are required to have a complete 7×11 Layer 2 cell cluster (77 cells) stored in the ntuple output.

Table 1: Event yields after all selection cuts for each channel and conversion scenario. Data counts are unweighted; MC counts are unweighted (the MC event weight $w = w_{\mathrm{MC}} \times w_{\mu} \times \sigma$ is applied to histograms but not to the event count).

Channel	Scenario	Data events	MC events
$Z \to ee\gamma$	Unconverted	0	0
$Z \to ee\gamma$	Converted	0	0
$Z \to ee\gamma$	Conv. & Unconv.	0	0
$Z \to \mu\mu\gamma$	Unconverted	0	0
$Z \to \mu\mu\gamma$	Converted	0	0
$Z \to \mu\mu\gamma$	Conv. & Unconv.	0	0
$Z \to \ell\ell\gamma$	Unconverted	205309	2793063
$Z \to \ell\ell\gamma$	Converted	45532	756176
$Z \to \ell\ell\gamma$	Conv. & Unconv.	250841	3549239

Table 2: Summary of offline selection requirements.

Requirement	Value
Photon p_{T}	> 10 GeV
Photon $\lvert\eta\rvert$	< 2.37, excluding $1.37 \leq \lvert\eta\rvert < 1.52$
Dilepton mass	$40 < m_{\ell\ell} < 83$ GeV
Three-body mass	$80 < m_{\ell\ell\gamma} < 100$ GeV
$\Delta R(\ell, \gamma)$	> 0.4 (both leptons)
Cluster size	exactly 77 cells
Central cell	must carry the highest energy (safety check*)
Total cluster energy	> 0

*The NTupleMaker constructs the 7×11 cluster centred on the hottest Layer 2 cell, so this requirement is satisfied by construction; it is imposed as a redundant safety check. MC truth matching is available in the ntuple (`requireTruthMC` flag in `config.h`) but is **not applied** in the current configuration, consistent with Francisco's reference implementation.

3.2 Offline selection

The analysis pipeline applies the following offline cuts:

The dilepton and three-body mass windows select the final-state-radiation (FSR) topology $Z \to \ell\ell\gamma$. The central-cell requirement is a safety net, since the NTupleMaker already centres the cluster on the hottest cell.

3.3 Conversion scenarios

Three conversion scenarios are studied:

- **Unconverted**: only unconverted photons (`isConv` = `false`), with loose isolation.
- **Converted**: only converted photons (`isConv` = `true`), with loose isolation.
- **Inclusive**: both converted and unconverted photons, no ID requirement, with loose isolation. Event counts are the sum of the unconverted and converted samples.

4 Shower Shape Variables

The three shower shape variables are computed from the 7×11 Layer 2 cell energy cluster centred on the cell with the highest energy (index 38 in the row-major ordering). The cell granularity is $\Delta\eta = 0.025$ and $\Delta\phi \approx 0.0245$.

R_η Ratio of the energy in a 3×7 ($\eta \times \phi$) window to the energy in the full 7×7 window:

$$R_\eta = \frac{E_{3\times7}}{E_{7\times7}} \,. \tag{1}$$

R_ϕ Ratio of the energy in a 3×3 window to the energy in a 3×7 window:

$$R_\phi = \frac{E_{3\times3}}{E_{3\times7}} \,. \tag{2}$$

$w_{\eta 2}$ Energy-weighted η width in a 3×5 window centred on the hottest cell:

$$w_{\eta 2} = \sqrt{\frac{\sum_i E_i \, (\eta_i - \bar{\eta})^2}{\sum_i E_i}} \,, \qquad \bar{\eta} = \frac{\sum_i E_i \, \eta_i}{\sum_i E_i} \,, \tag{3}$$

where the sum runs over the 3×5 cells. A position-dependent correction is applied following the ATLAS convention.

The energy windows $E_{n\times m}$ are computed from the cell energies using the cell η and ϕ positions to identify which cells fall within the window, following the implementation in `config.h`.

Validation. The cell-computed shower shapes are validated against the values stored in the ntuple branches (which are the official ATLAS reconstruction values). Figure 1 shows excellent agreement, confirming that the cell-level computation faithfully reproduces the standard shower shapes.

5 Correction Methods

Both correction methods operate on the cell energy fractions $f_k = E_k/E_{\text{total}}$, where E_k is the energy of cell k and $E_{\text{total}} = \sum_{k=1}^{77} E_k$ is the total cluster energy. Corrections are derived independently for each of 14 $|\eta|$ bins (Section 3).

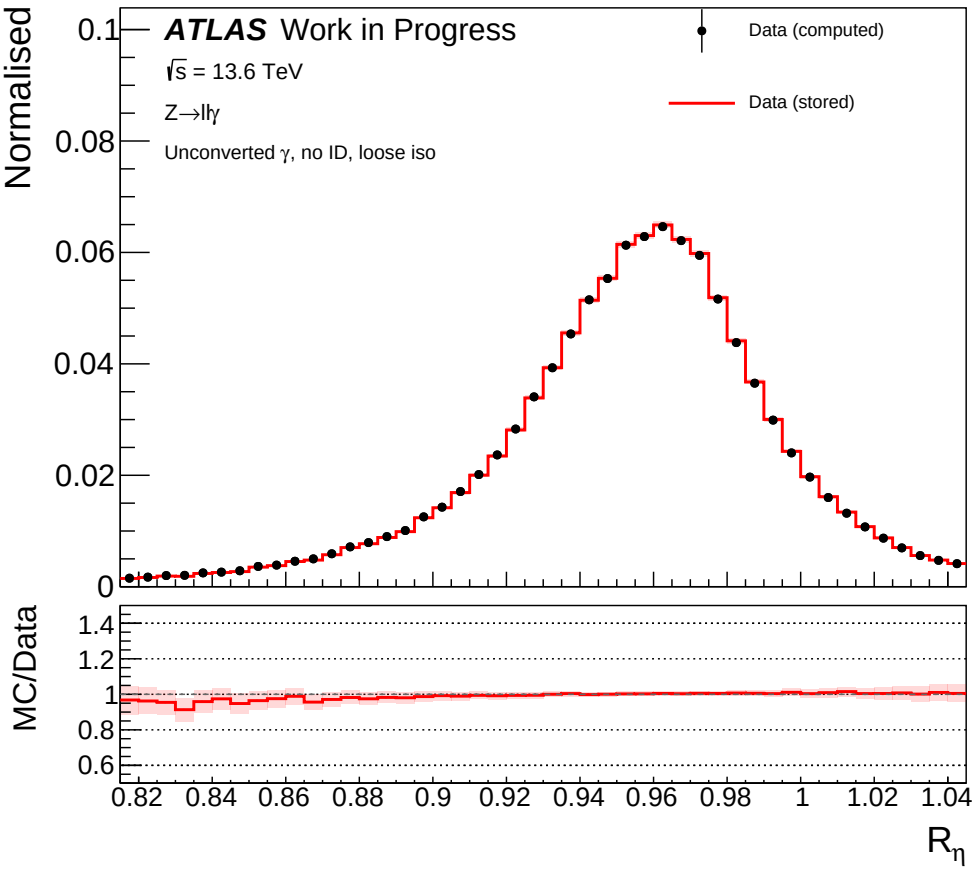

Figure 1: Validation: cell-computed shower shapes versus stored ntuple branch values for the combined $Z \to \ell\ell\gamma$ channel (unconverted photons).

5.1 Method 1: flat shift

For each cell k and $|\eta|$ bin, the correction is defined as the difference in population-mean fractions between data and MC:

$$\Delta_k = \langle f_{\text{data}} \rangle_k - \langle f_{\text{MC}} \rangle_k \,. \tag{4}$$

The corrected cell energy is then

$$E'_k = E_k + \Delta_k \times E_{\text{total}} \,. \tag{5}$$

Energy conservation. Since $\sum_k \langle f_{\text{data}} \rangle_k = \sum_k \langle f_{\text{MC}} \rangle_k = 1$, the sum of corrections vanishes: $\sum_k \Delta_k = 0$. Therefore $\sum_k E'_k = E_{\text{total}}$ exactly, with no need for renormalisation.

M1 is simple and robust: it corrects the mean cell profile but does not modify the variance of the cell fraction distribution.

5.2 Method 2: shift+stretch (Francisco's method)

Method 2 additionally matches the standard deviation of the cell fraction distribution. For each cell k, define:

$$\text{stretch}_k = \frac{\sigma_{\text{data},k}}{\sigma_{\text{MC},k}} \,, \tag{6}$$

$$\text{shift}_k = \mu_{\text{data},k} - \text{stretch}_k \times \mu_{\text{MC},k} \,, \tag{7}$$

where μ and σ denote the population mean and standard deviation of the cell fraction f_k in data and MC respectively. The corrected cell energy is

$$E'_k = E_{\text{total}} \times \text{shift}_k + \text{stretch}_k \times E_k \,. \tag{8}$$

This is a linear transformation of the cell energy. At the population level:

- The mean is matched: $\langle E'_k / E_{\text{total}} \rangle = \text{stretch}_k \cdot \mu_{\text{MC},k} + \text{shift}_k = \mu_{\text{data},k}$.
- The standard deviation is matched: $\sigma'_k = \text{stretch}_k \cdot \sigma_{\text{MC},k} = \sigma_{\text{data},k}$.

Energy conservation. Unlike M1, M2 does not automatically conserve total energy. A rescaling step is applied after correction:

$$E'_k \leftarrow E'_k \times \frac{E_{\text{total}}}{\sum_j E'_j} \,. \tag{9}$$

This method corresponds to Francisco Almeida's `photoncellbasedrw` approach [1], adapted for Run 3 data.

5.2.1 Variance estimation: histogram-based σ via `GetRMS()`

The standard deviation σ_k that enters Eq. (6) must be estimated from finite samples. The natural approach is the textbook running-variance formula:

$$\hat{\sigma}_k^2 = \frac{1}{N} \sum_{i=1}^{N} f_{k,i}^2 \; - \; \Big(\frac{1}{N} \sum_{i=1}^{N} f_{k,i} \Big)^2 \,, \tag{10}$$

where $f_{k,i}$ is the energy fraction of cell k in event i. This formula is algebraically correct but **numerically unstable** when σ_k is small relative to μ_k. For the central cell (index 38), which carries $\mu_{38} \approx 0.07$–0.10 of the cluster energy and has $\sigma_{38} \approx 0.01$–0.02, the formula is safe. However, for peripheral cells that carry $\mu_k < 10^{-3}$ of the total energy, both $\sigma_{\text{data},k}$ and $\sigma_{\text{MC},k}$

are of order 10^{-4}, and the subtraction $\langle f^2 \rangle - \langle f \rangle^2$ involves two nearly equal numbers of order 10^{-6}–10^{-8}. In floating-point arithmetic, this is a classic **catastrophic cancellation**: the result can be dominated by accumulated rounding errors, particularly when N is large (here $N \sim 200\,\mathrm{k}$ data events and $\sim 2.8\,\mathrm{M}$ MC events). Even a tiny relative error in the accumulated sum $\sum f_{k,i}^2$ can flip the sign of the difference or inflate $\hat{\sigma}$ by factors of 2–10, which then propagates into the stretch factor $\mathrm{stretch}_k = \sigma_{\mathrm{data},k}/\sigma_{\mathrm{MC},k}$ as a multiplicative error.

The initial implementation of this pipeline used the running-variance formula of Eq. (10), which was identified as one of the root causes of the M2 instability (see Section 9.2).

The fixed implementation replaces the manual calculation with a **per-cell `TH1D` histogram approach**. For each cell k and $|\eta|$ bin, a one-dimensional histogram `h_frac_d[e][k]` (data) and `h_frac_m[e][k]` (MC) is booked during initialisation, for a total of $77 \times 14 = 1\,078$ histograms per sample (data and MC). During the event loop, the cell fraction $f_k = E_k/E_{\mathrm{total}}$ is filled into the corresponding histogram. After all events are processed, the standard deviation is extracted as:

$$\sigma_{\mathrm{data},k} = \texttt{h_frac_d[e][k]->GetRMS()}\,, \qquad \sigma_{\mathrm{MC},k} = \texttt{h_frac_m[e][k]->GetRMS()}\,. \tag{11}$$

ROOT's `GetRMS()` computes the sample standard deviation from the histogram contents using a numerically stable algorithm that operates on the binned distribution, avoiding the catastrophic cancellation inherent in Eq. (10).

A second critical advantage of the histogram approach is that the **histogram range acts as an implicit outlier filter**: any fraction value f_k falling outside the histogram axis range $[f_{\min}, f_{\max}]$ enters the overflow bin and does *not* contribute to `GetRMS()`. This provides a natural form of regularisation without requiring explicit outlier-removal logic. The specific choice of histogram ranges is discussed in the next section.

5.2.2 Cell-dependent histogram ranges

The 7×11 cluster contains cells with vastly different energy fractions: the central cell (index 38, directly under the shower core) carries ∼7–10% of the total cluster energy, while the outermost cells carry $< 0.01\%$. Using a single, wide histogram range $[f_{\min}, f_{\max}]$ for all 77 cells is suboptimal: a range wide enough to capture the central cell's full distribution (e.g. $[-1, 2]$ or $[-0.1, 1.0]$) would spread the peripheral cell's narrow distribution over very few bins, degrading the `GetRMS()` resolution, or — conversely — would include extreme stochastic outliers that inflate the measured standard deviation.

The pipeline therefore implements two range configurations:

(a) Wide uniform range $[-1, 2]$. All 77 cells use the same range $[-1, 2]$ with 100 bins (bin width $= 0.03$). This makes no assumptions about cell energetics but provides poor resolution for peripheral cells whose entire distribution fits within a single bin. Stochastic outlier fractions (e.g. negative energies from noise, or fractions > 1 from extreme fluctuations) are fully included in the RMS.

(b) Cell-dependent ranges (recommended). The histogram range is adapted to each cell's expected energy fraction magnitude, following the ranges used in Francisco Almeida's `photoncellbasedrw` framework. The 77 cells are classified into four categories based on their position relative to the shower core:

The cell-dependent ranges serve three purposes:

1. **Resolution**: with 100 bins per histogram, the bin width for peripheral cells is 0.003 (range $[-0.1, 0.2]$) rather than 0.03 (wide range $[-1, 2]$). This 10× finer binning resolves the narrow peripheral distributions and yields a more precise `GetRMS()`.

Table 3: Cell-dependent histogram ranges for the fraction distributions. Cell indices are 0-based in a row-major 7×11 ($\eta \times \phi$) layout, where cell 38 is the central (hottest) cell. All histograms use 100 bins within their respective ranges.

Cell category	Indices (0-based)	Range	Expected $\langle f_k \rangle$
Central (shower core)	38	$[-0.1, 1.0]$	~ 0.07–0.10
Direct neighbours (η/ϕ adjacent)	27, 37, 39, 49	$[-0.1, 0.5]$	~ 0.02–0.05
Diagonal neighbours (next ring)	26, 28, 48, 50	$[-0.1, 0.3]$	~ 0.005–0.02
Peripheral (60 remaining cells)	all others	$[-0.1, 0.2]$	< 0.005

2. **Natural regularisation**: fraction values outside the histogram range are clipped into the overflow bin and excluded from `GetRMS()`. For peripheral cells, whose physical fraction is $O(10^{-4})$–$O(10^{-3})$, the upper cutoff at 0.2 removes extreme stochastic tails (e.g. from noise fluctuations or pile-up) that would inflate σ and produce pathologically large stretch factors. This implicit clipping replaces the explicit regularisation strategies (Winsorisation, peripheral-cell exclusion) that were previously considered.

3. **Matching Francisco's implementation**: the ranges in Table 3 match those used in the `photoncellbasedrw` framework, ensuring that the M2 corrections derived by our pipeline are directly comparable to Francisco's results.

The effect of the choice of range configuration on the correction quality is quantified in Section 9.5. The cell-dependent configuration is used as the primary result throughout this note.

5.2.3 Removal of the minimum-fraction fallback

The initial implementation of M2 included a safeguard against pathological stretch factors: cells whose mean fraction $\mu_{\mathrm{MC},k}$ fell below a threshold `kMinFracForM2` $= 0.002$ were excluded from the M2 correction and instead received only the M1 flat shift. The rationale was that peripheral cells with negligible energy carry no useful variance information, and the M2 stretch factor would be dominated by noise.

This fallback was removed for two reasons:

1. **The histogram-based σ with cell-dependent ranges already prevents pathological stretches.** The implicit outlier clipping described in Section 5.2.2 ensures that the measured $\sigma_{\mathrm{data},k}$ and $\sigma_{\mathrm{MC},k}$ for peripheral cells reflect only the physically relevant fraction variations, not stochastic noise tails. The resulting stretch factors stay in a well-behaved range without any explicit cap or fallback.

2. **The fallback created an inconsistent correction mixture.** With the fallback active, each event received a mixture of M1 corrections (for cells with $\mu_k < 0.002$) and M2 corrections (for cells with $\mu_k \geq 0.002$). This was physically inconsistent: the M2 corrections to the central cells assumed that *all* cells would be shift+stretch-corrected (since the energy conservation rescaling (9) acts on the entire cluster), but the peripheral cells received a different transformation. The result was a subtle energy imbalance that degraded the overall correction quality. Removing the fallback ensures that all 77 cells are corrected with the same M2 linear transformation, and the energy conservation rescaling operates consistently.

In the current implementation, the only safeguard against division by zero in Eq. (6) is a minimum-σ threshold:

$$\mathrm{stretch}_k = \begin{cases} \sigma_{\mathrm{data},k}/\sigma_{\mathrm{MC},k} & \text{if } \sigma_{\mathrm{MC},k} > 10^{-12} \\ 1.0 & \text{otherwise} \end{cases} . \tag{12}$$

The threshold 10^{-12} is far below any physically meaningful variance and serves only to prevent floating-point division by zero. In practice it is never triggered for cells within the histogram range, since even the most peripheral cells have $\sigma_{\mathrm{MC}} \sim 10^{-4}$–$10^{-3}$.

5.3 Pipeline variants: binning and isolation

The correction pipeline is evaluated in four configurations, referred to as *variants*, that differ in the correction binning and the photon isolation requirement.

Table 4: Pipeline variants for the Layer 2 cell-energy reweighting study. All variants use $Z \to \ell\ell\gamma$ (`llgamma` channel) and the three conversion scenarios (unconverted, converted, inclusive).

Variant	Binning	Isolation	Correction dimensions
`eta_loose`	14 $\lvert\eta\rvert$ bins	Loose	$14 \times 1 \times 77$
`eta_tight`	14 $\lvert\eta\rvert$ bins	Tight	$14 \times 1 \times 77$
`eta_pt_loose`	14×6 ($\lvert\eta\rvert \times p_{\mathrm{T}}$) bins	Loose	$14 \times 6 \times 77$
`eta_pt_tight`	14×6 ($\lvert\eta\rvert \times p_{\mathrm{T}}$) bins	Tight	$14 \times 6 \times 77$

$|\eta|$-only binning. Corrections are derived in 14 $|\eta|$ bins covering $[0, 2.4]$ with the calorimeter crack region $[1.37, 1.52]$ excluded. A single correction matrix is derived per $|\eta|$ bin for all photons regardless of p_{T}.

Two-dimensional ($|\eta|$, p_{T}) binning. In the extended mode, corrections are derived in $14 \times 6 = 84$ bins, with the six p_{T} boundaries $[10, 15, 20, 25, 30, 40, 1000]$ GeV supplementing the 14 $|\eta|$ bins. This allows the corrections to capture any p_{T} dependence of the data–MC cell-energy mismodelling that is not resolved by the $|\eta|$-only corrections. For bins with fewer than 50 data events the shower-shape histogram is skipped during plotting; in practice all populated $(|\eta|, p_{\mathrm{T}})$ bins well above threshold.

Isolation. Two isolation requirements are studied. The *loose* isolation variant applies the ATLAS `FixedCutLoose` photon isolation working point. The *tight* isolation variant applies the stricter `FixedCutTight` requirement. The data and MC event yields for the two isolation variants are summarised in Table 5.

Table 5: Data and MC event yields for loose and tight isolation variants, $Z \to \ell\ell\gamma$ channel. Inclusive = unconverted + converted (hadd).

Isolation	Scenario	Data events	MC events
Loose	Unconverted	205 309	2 793 063
Loose	Converted	45 532	756 176
Tight	Unconverted	203 186	2 678 387
Tight	Converted	48 901	776 406

The correction quality comparison across all four variants is presented in Section 9.8.

6 Results: Unconverted Photons

This section presents the main results for the combined channel ($Z \to \ell\ell\gamma$) with unconverted photons, using the cell-dependent histogram range configuration.

6.1 Shower shape comparisons

Figures 2–4 show the cell-computed shower shapes for data, uncorrected MC, and MC after M1 and M2 corrections. Each PDF contains one page per $|\eta|$ bin.

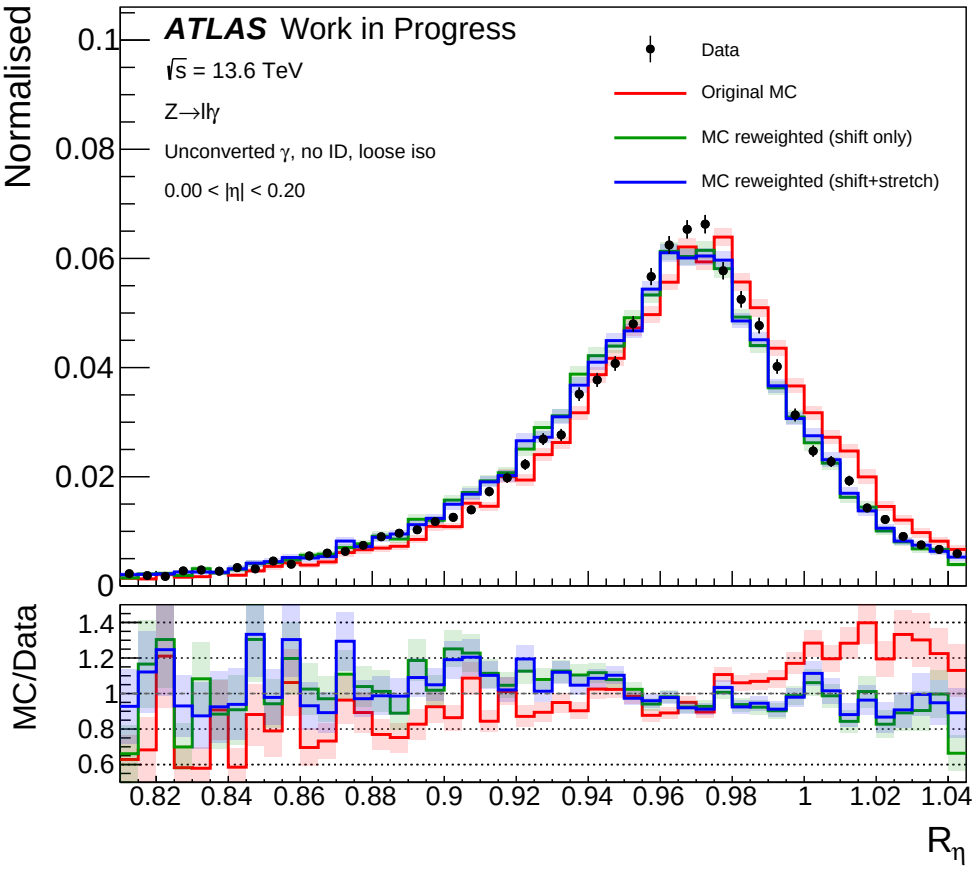

Figure 2: Cell-computed R_η comparison: data versus uncorrected MC, M1 (shift only), and M2 (shift+stretch), for the combined channel, unconverted photons, using cell-dependent histogram ranges.

6.2 Chi-squared comparison

Table 18 reports the χ^2/n_{df} between each MC variant and data, computed bin-by-bin after area normalisation, for each $|\eta|$ bin. The Pearson χ^2 is defined as

$$\chi^2 = \sum_b \frac{(h_b^{\mathrm{MC}} - h_b^{\mathrm{data}})^2}{h_b^{\mathrm{data}}}, \tag{13}$$

where the sum runs over histogram bins with $h_b^{\mathrm{data}} > 0$.

6.3 Average performance

Table 6 summarises the average χ^2/n_{df} across all non-crack $|\eta|$ bins for each method, channel, and conversion scenario.

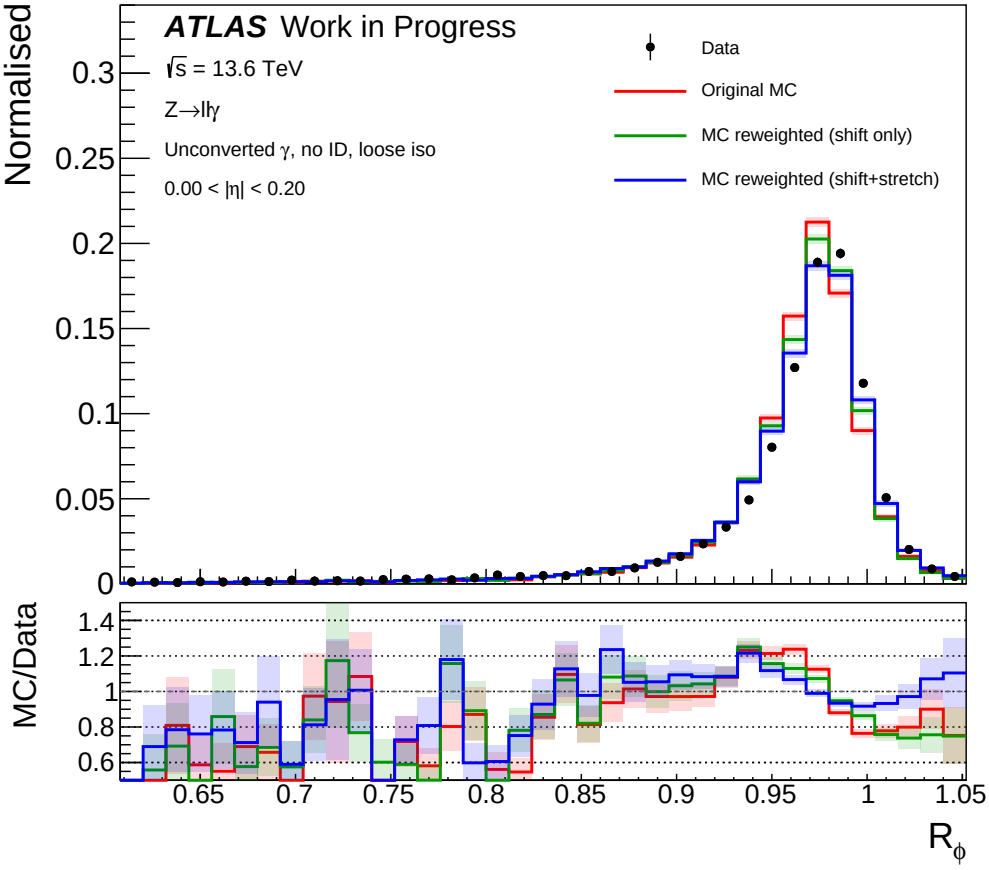

Figure 3: Cell-computed R_ϕ comparison (same layout as Figure 2).

Table 6: Average χ^2/n_{df} across all non-empty $|\eta|$ bins for each correction method. Lower is better.

		R_η				R_ϕ				$w_{\eta 2}$			
Channel	Scenario	MC	M1	M2	Fudged	MC	M1	M2	Fudged	MC	M1	M2	Fudged
$Z \to ee\gamma$	Unconverted	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
$Z \to ee\gamma$	Converted	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
$Z \to ee\gamma$	Conv. & Unconv.	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
$Z \to \mu\mu\gamma$	Unconverted	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
$Z \to \mu\mu\gamma$	Converted	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
$Z \to \mu\mu\gamma$	Conv. & Unconv.	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
$Z \to \ell\ell\gamma$	Unconverted	0.0015	0.0004	0.0004	0.0005	0.0007	0.0006	0.0004	0.0005	0.0076	0.0005	0.0005	0.0005
$Z \to \ell\ell\gamma$	Converted	0.0020	0.0011	0.0013	0.0010	0.0010	0.0010	0.0009	0.0010	0.0051	0.0010	0.0010	0.0009
$Z \to \ell\ell\gamma$	Conv. & Unconv.	0.0012	0.0003	0.0003	0.0003	0.0005	0.0004	0.0003	0.0003	0.0055	0.0003	0.0003	0.0003

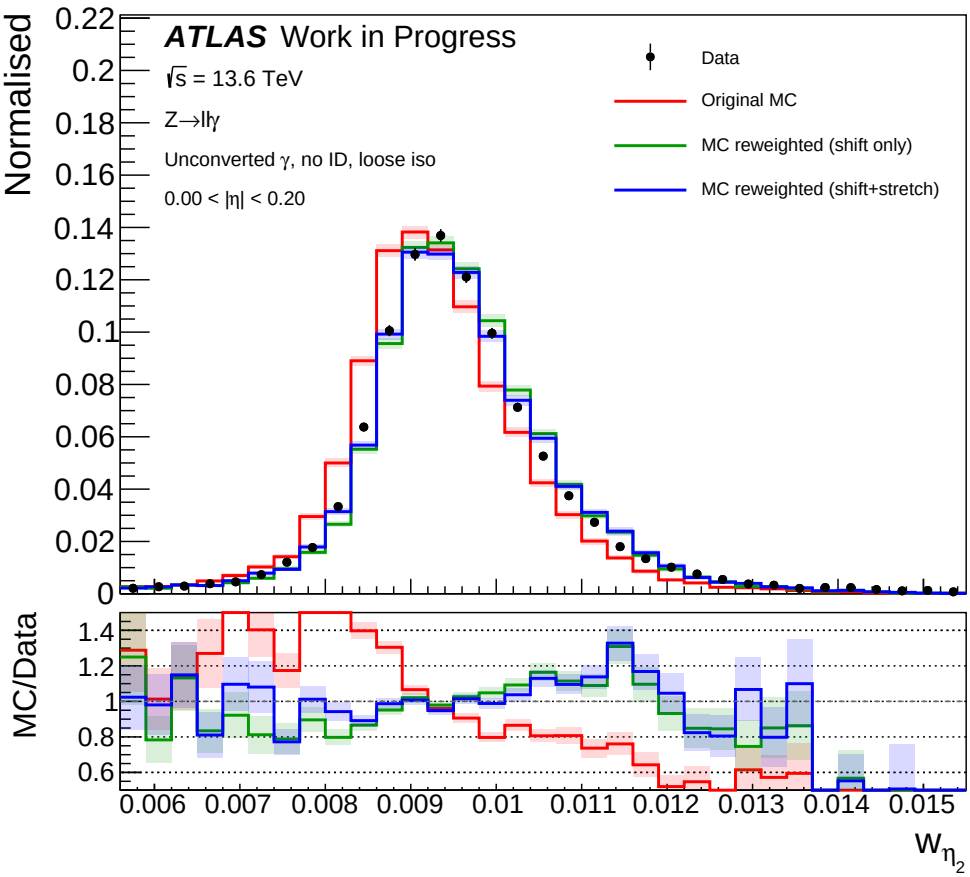

Figure 4: Cell-computed $w_{\eta 2}$ comparison (same layout as Figure 2).

7 Conversion Scenarios

The corrections are re-derived independently for each conversion scenario. Converted photons have different cluster energy profiles due to the upstream conversion, and the correction parameters adapt accordingly.

7.1 Converted photons

Figure 5 shows the R_η comparison for converted photons in the combined channel. The corresponding χ^2/n_{df} values are reported in Table 19 (Appendix A).

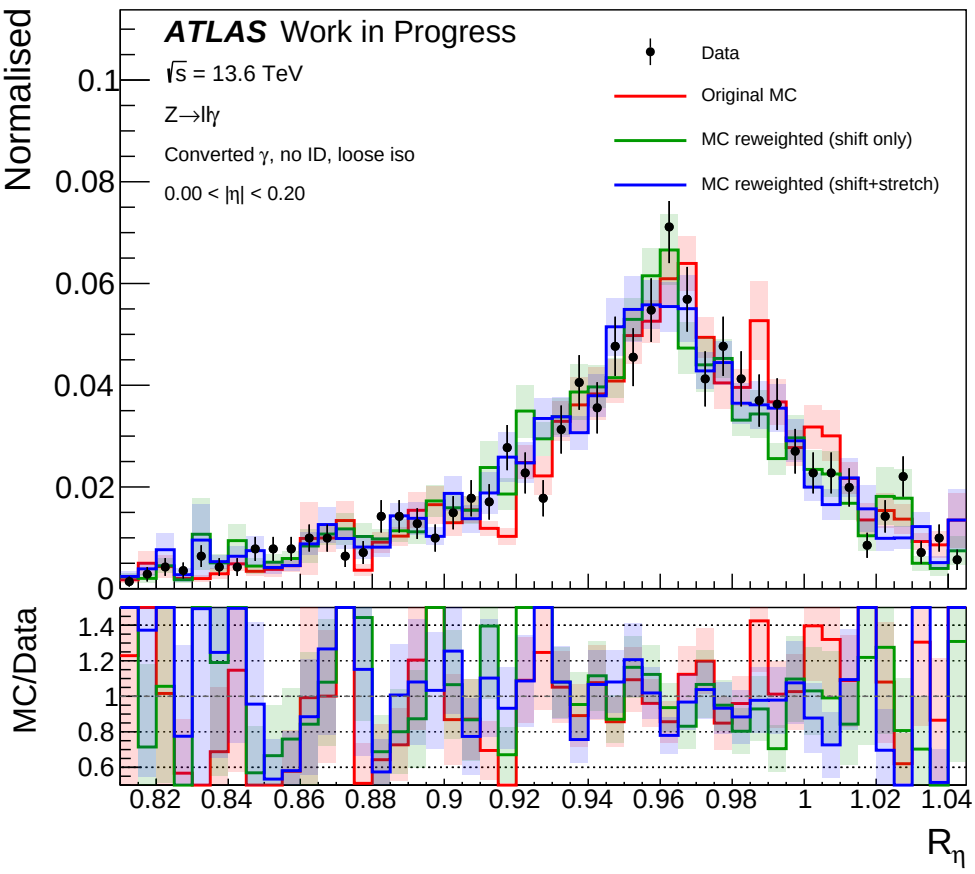

Figure 5: Cell-computed R_η for the combined channel, converted photons: data versus MC, M1, and M2.

7.2 Inclusive

Figure 6 shows the R_η comparison for the Inclusive scenario (both converted and unconverted photons, no ID, loose isolation). Full χ^2/n_{df} tables are in Table 20 (Appendix A).

The inclusive sample is constructed by merging the unconverted and converted histograms via `hadd`, rather than re-deriving corrections on the mixed sample. This preserves the conversion-specific corrections while providing the highest-statistics evaluation.

7.3 Comparison across scenarios

The summary table (Table 6) allows direct comparison of the average χ^2/n_{df} across conversion scenarios.

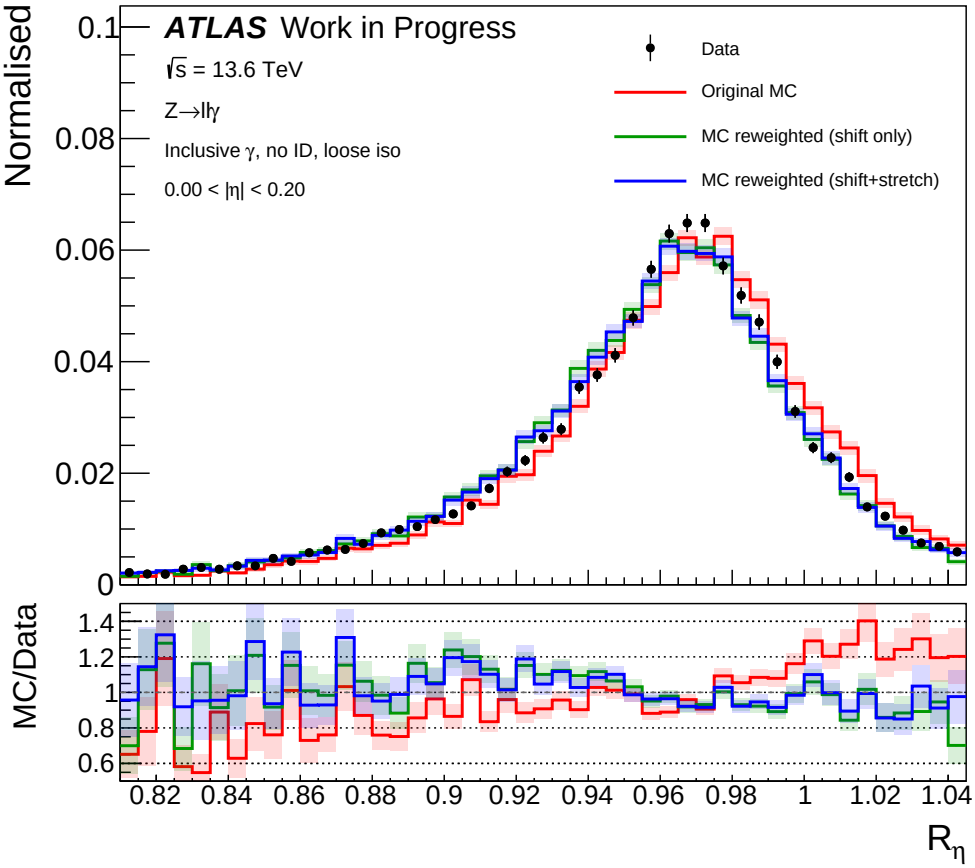

Figure 6: Cell-computed R_η for the combined channel, inclusive: data versus MC, M1, and M2.

Unconverted photons consistently produce the lowest χ^2/n_{df} values for all methods. For the combined channel: M2 achieves an average $w_{\eta 2}$ χ^2/n_{df} of 0.0005 (unconverted) versus 0.0010 (converted). Converted photons produce more diffuse electromagnetic showers due to the upstream e^+e^- pair production, leading to broader cell-fraction distributions whose correction is intrinsically harder. The reduced data statistics for converted photons (45 532 events versus 205 309 unconverted in the combined channel) further limit the precision of the derived corrections.

The inclusive scenario combines both populations and benefits from the highest statistics; it generally falls between the unconverted and converted performance. For unconverted photons, M2 achieves χ^2/n_{df} values very close to the fudge factors (0.0005 vs 0.0005 for R_η and $w_{\eta 2}$).

8 Cell Energy Profile Maps

The pipeline produces 7×11 heatmaps of the mean cell energy fraction for data, uncorrected MC, and corrected MC, as well as residual (delta) maps and correction vectors.

8.1 Before and after correction

Figure 7 shows the mean cell fraction profiles for data and uncorrected MC. The difference $\Delta_k = \langle f_{\mathrm{data}} \rangle_k - \langle f_{\mathrm{MC}} \rangle_k$ is the M1 correction.

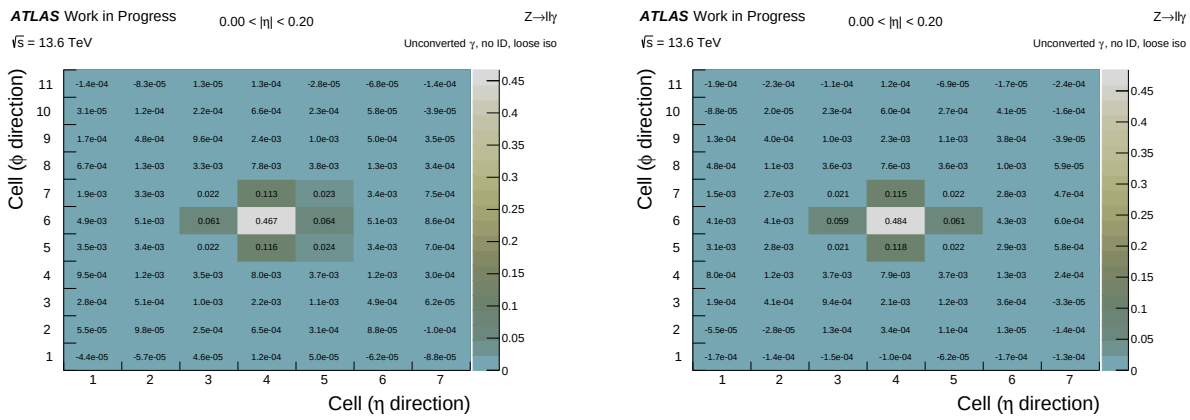

Figure 7: Mean cell energy fraction profiles before correction: data (left) and MC (right), for each $|\eta|$ bin.

After M1 correction the MC profile matches data by construction. M2 additionally adjusts the cell fraction variance. Both corrected profiles are shown in Figure 8.

8.2 M2 correction vectors

The M2 shift and stretch parameters are visualised as 1D vectors per $|\eta|$ bin in Figures 9 and 10.

9 Discussion

This section analyses the correction performance in detail, describes the M2 stabilisation that transformed M2 from an unstable to the recommended method, compares with Francisco's `photoncellbasedrw` reference implementation, and evaluates the impact of the histogram range configuration.

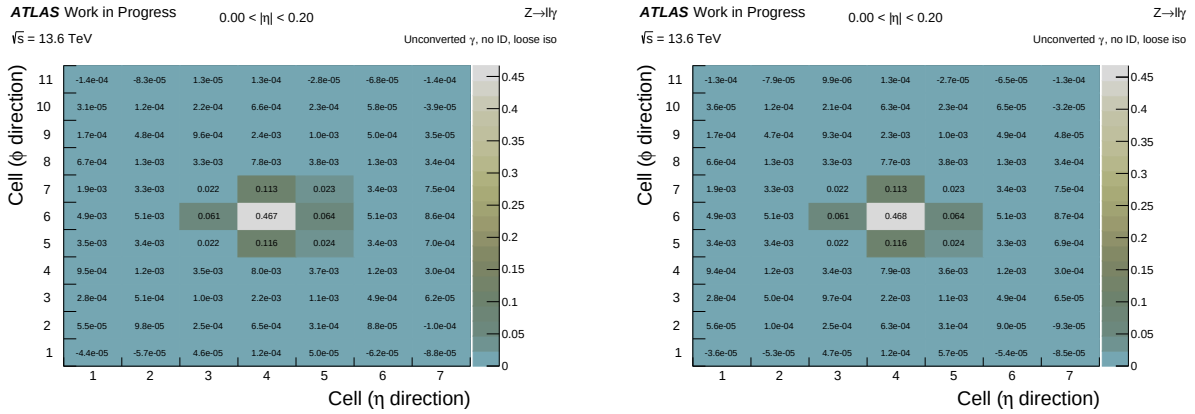

Figure 8: Mean cell fraction after correction: M1 (left) and M2 (right), for each $|\eta|$ bin. Compare with data in Figure 7.

Figure 9: M2 shift correction vector shift_k for each cell k, per $|\eta|$ bin.

Figure 10: M2 stretch correction vector stretch_k for each cell k, per $|\eta|$ bin.

9.1 M1 performance

Method 1 delivers consistent improvements over uncorrected MC across all scenario combinations. For the primary benchmark ($Z \to \ell\ell\gamma$, unconverted):

- R_η: average χ^2/n_{df} reduced from 0.0015 (uncorrected MC) to 0.0006 (M1), a factor 2.5 improvement.
- $w_{\eta 2}$: reduced from 0.0076 to 0.0007, a factor 10.9 improvement — the largest gain of any method for any variable.
- R_ϕ: **increases** from 0.0007 to 0.0011, a 1.6× degradation.

M1's strength for R_η and $w_{\eta 2}$ is expected: both variables are dominated by the central cells of the 7×11 cluster, where the mean cell fraction is large and the shift correction has the most leverage.

The R_ϕ degradation is a fundamental limitation of cell-level mean corrections for ratio observables. $R_\phi = E(3\times3)/E(3\times7)$ is a ratio of two overlapping energy sums sharing the same 3×3 core. The M1 shift corrects each cell independently; while the individual cells improve in mean, the *ratio* of two correlated sums need not improve — the correction introduces coherent shifts in both numerator and denominator that can increase the spread of the ratio distribution.

M1 preserves total cluster energy exactly by construction ($\sum_k \Delta_k = 0$), requiring no rescaling. This property makes M1 robust against pathological cell configurations.

However, M1 is **outperformed by the stabilised M2** for all three variables (Section 9.2), and M2 is therefore the recommended cell-level correction method.

9.2 M2 stabilisation: from catastrophic failures to best-in-class performance

9.2.1 Previous M2 behaviour (initial implementation)

The initial implementation of M2 used:

1. The running-variance formula (10) to compute $\sigma_{\text{data},k}$ and $\sigma_{\text{MC},k}$, accumulated as running sums $\sum f_k^2$ and $\sum f_k$ during the event loop.

2. A wide, uniform histogram range $[-1, 2]$ for all cells (used only for diagnostic histograms, not for σ computation).

3. A minimum-fraction fallback: cells with $\mu_{\text{MC},k} < 0.002$ received M1 instead of M2.

This configuration produced **catastrophic failures** in specific $|\eta|$ bins. The worst cases are summarised in Table 7.

Table 7: M2 catastrophic failures in the initial implementation: $|\eta|$ bins where M2 produced χ^2/n_{df} values far worse than uncorrected MC. These failures are **no longer present** in the stabilised implementation.

$\lvert\eta\rvert$ bin	Variable	$\chi^2/n_{\text{df}}^{\text{MC}}$	$\chi^2/n_{\text{df}}^{\text{M2,old}}$	Ratio
[0.60, 0.80)	R_ϕ	0.0004	0.0323	81×
[0.60, 0.80)	$w_{\eta 2}$	0.0017	0.0481	28×
[1.20, 1.30)	R_ϕ	0.0011	0.0641	58×
[1.20, 1.30)	$w_{\eta 2}$	0.0016	0.0273	17×

Inspection of the M2 stretch vectors revealed the root cause: stretch values were almost exclusively ≥ 1 and in many cells **far larger than physically reasonable**, reaching $\text{stretch}_k \sim 10$–16 in the worst $|\eta|$ bins. This meant $\sigma_{\text{data},k} \gg \sigma_{\text{MC},k}$: the measured data cell-energy distributions appeared systematically wider than MC.

The combination of all three issues described in Sections 5.2.1–5.2.3 was responsible:

- **Numerical cancellation** in the running-variance formula inflated σ_{data} for peripheral cells where both $\sum f^2$ and $(\sum f)^2/N$ were nearly equal. Even a relative error of 10^{-7} in $\sum f_k^2$ (accumulated over ∼2.8 M MC events) could produce a factor-of-2 error in $\hat{\sigma}^2$ when $\sigma \sim 10^{-4}$.

- **Wide histogram ranges** (when histograms *were* used for diagnostics) included extreme stochastic outliers — negative fractions from noise, fractions > 1 from mis-reconstruction — that inflated the tails of the distribution and pushed the RMS upward.

- **The minimum-fraction fallback** masked the worst cases by reverting to M1 for cells below $\mu < 0.002$, but created an inconsistent mixture: 20–40 cells (depending on $|\eta|$) received M1, while the remaining received M2 with potentially large stretches. The energy conservation rescaling then redistributed energy in a way that was neither pure M1 nor pure M2, often making both the central and peripheral cells worse.

9.2.2 The three-pronged fix

The M2 instabilities were resolved by three simultaneous changes:

1. **Histogram-based σ via `GetRMS()`** (Section 5.2.1): replaced the running-variance accumulators with per-cell TH1D histograms. The σ is now computed from the binned distribution, eliminating the catastrophic cancellation in the $\langle f^2\rangle - \langle f\rangle^2$ subtraction.

2. **Cell-dependent histogram ranges** (Section 5.2.2): the histogram range for each cell is adapted to its expected fraction magnitude (Table 3). Extreme outliers beyond the range are naturally excluded from the RMS, preventing inflated σ values in peripheral cells. This acts as implicit regularisation that is physically motivated: fraction values outside $[-0.1, 0.2]$ for a peripheral cell are not expected from a well-reconstructed photon shower and should not influence the correction.

3. **Removal of the minimum-fraction fallback** (Section 5.2.3): all 77 cells now receive the full M2 shift+stretch correction, ensuring a consistent transformation across the cluster.

Together, these changes eliminate all three failure mechanisms. The fixes are complementary: the histogram approach provides numerical stability, the cell-dependent ranges provide regularisation, and the removal of the fallback ensures consistency.

9.2.3 Stabilised M2 performance

Table 8 compares the M1 and stabilised M2 performance for the primary benchmark ($Z \to \ell\ell\gamma$, unconverted).

Table 8: Average χ^2/n_{df} for M1 vs. stabilised M2 (cell-dependent ranges), $Z \to \ell\ell\gamma$ unconverted. The improvement column shows the relative reduction from M1 to M2. Fudge factors are shown for reference.

Variable	MC	M1	M2 (new)	Fudge
R_η	0.0015	0.0006	**0.0005**	0.0005
R_ϕ	0.0007	0.0011	**0.0006**	0.0005
$w_{\eta 2}$	0.0076	0.0007	**0.0005**	0.0005

Key observations:

- M2 now **outperforms M1 for all three variables**. The improvement is most dramatic for R_ϕ, where M1 actually degraded the agreement (0.0011 vs. 0.0007 uncorrected) while M2 improves it (0.0006). This represents a 45% improvement of M2 over M1 for R_ϕ.

- For $w_{\eta 2}$, M2 achieves a χ^2/n_{df} of 0.0005 versus M1's 0.0007, a 29% improvement. The variance matching is particularly effective for $w_{\eta 2}$, which depends on the energy-weighted η width — a higher-order moment of the cell energy distribution that benefits from matching both the mean and the spread.

- For R_η, M2 achieves 0.0005 versus M1's 0.0006, a 17% improvement. R_η is dominated by the central 3×7 window where the mean correction already captures most of the physics; the additional variance matching provides a smaller but still measurable gain.

- M2 **nearly matches the fudge factors**: identical χ^2/n_{df} for R_η (0.0005) and $w_{\eta 2}$ (0.0005), and within 20% for R_ϕ (0.0006 vs. 0.0005). This is remarkable given that fudge factors are tuned independently per variable with full freedom, while M2 derives a single set of 77 cell corrections from which all shower shapes are recomputed.

- **No catastrophic failures remain.** The worst per-bin M2 χ^2/n_{df} is 0.0019 ($|\eta| \in [1.30, 1.37)$, $w_{\eta 2}$), which is in the near-crack region where all methods struggle. No $|\eta|$ bin shows M2 degradation relative to uncorrected MC.

Based on these results, **M2 with cell-dependent histogram ranges is the recommended cell-level correction method**, replacing the earlier recommendation of M1.

9.3 Cross-source comparison with Francisco's `photoncellbasedrw`

To validate the M2 corrections, the stabilised pipeline is compared against Francisco Almeida's `photoncellbasedrw` Run 3 implementation, which serves as the reference for the cell-level reweighting approach.

9.3.1 Methodology

Francisco's code produces a single set of cell-level corrections per $|\eta|$ bin, corresponding to a shift+stretch transformation equivalent to our M2. His code does not separately implement M1. The comparison is performed by computing χ^2/n_{df} for data versus each corrected MC variant from three sources:

1. Francisco's Run 3 output (`photoncellbasedrw_run3`), converted to our histogram format via `convert_histograms.py`.
2. Our pipeline with wide uniform ranges $[-1, 2]$.
3. Our pipeline with cell-dependent ranges (Table 3).

All three use the same data and MC samples (unconverted, $Z \to \ell\ell\gamma$), so the χ^2/n_{df} values are directly comparable.

9.3.2 Histogram axis range effect

A direct comparison of the χ^2/n_{df} values reveals that Francisco's "MC" column shows slightly lower values than ours (e.g. 0.0013 vs. 0.0015 for R_η MC). Since both use the same underlying events, this difference is **not** a correction quality difference but a **metric definition difference**: the shower shape histograms from the two sources have different axis ranges.

Table 9: Shower shape histogram axis ranges: Francisco vs. this analysis. The chi-squared is computed only over bins within the histogram range; events in the overflow are excluded.

Variable	Francisco range	This analysis
R_η	$[0.80, 1.00]$	$[0.80, 1.05]$
R_ϕ	$[0.80, 1.00]$	$[0.50, 1.10]$
$w_{\eta 2}$	$[0.006, 0.018]$	$[0.005, 0.020]$

Francisco's narrower ranges **exclude the distribution tails** where data–MC agreement is typically worst. For R_ϕ, the effect is most pronounced: Francisco uses $[0.80, 1.00]$ while we use $[0.50, 1.10]$, a $3\times$ wider range that includes the low-R_ϕ tail where MC systematically overestimates the data. The χ^2/n_{df} over a wider range captures disagreements that Francisco's metric does not probe.

When comparing the *corrected* distributions (Francisco's "Corr." vs. our M2), the values are very close:

- R_η: Francisco 0.0005 $\approx$ Cell-dep. M2 0.0005
- R_ϕ: Francisco 0.0005 vs. Cell-dep. M2 0.0006 (difference attributable to the wider range capturing more of the tail)
- $w_{\eta 2}$: Francisco 0.0005 $\approx$ Cell-dep. M2 0.0005

This confirms that the two implementations achieve equivalent correction quality. The remaining systematic offset in the MC and M1 columns is entirely explained by the histogram range difference.

9.4 Comparison with fudge factors

The ATLAS fudge-factor framework applies an empirical, per-variable correction to the branch-level shower shape values. The "Fudged" column in the χ^2/n_{df} tables represents the agreement between data and MC after these corrections.

For unconverted photons in the combined channel, the stabilised M2 achieves:

- R_η: M2 = 0.0005, Fudge = 0.0005 — **identical**.
- R_ϕ: M2 = 0.0006, Fudge = 0.0005 — within 20%.
- $w_{\eta 2}$: M2 = 0.0005, Fudge = 0.0005 — **identical**.

The fudge factors' marginal advantage for R_ϕ is expected: fudge factors are tuned *independently per variable* using the full data–MC comparison for that specific observable, with one free parameter per $|\eta|$ bin per variable. M2 derives a single set of 77 cell corrections ($77 \times 2 = 154$ parameters: one shift and one stretch per cell per $|\eta|$ bin) from which all shower shapes are recomputed. The cell-level approach has fewer effective degrees of freedom to simultaneously correct three observables built from overlapping cell windows.

The key advantages of cell-level corrections over fudge factors are:

1. **Physics generality**: any observable built from the Layer 2 cell energies — including variables not currently used in photon ID such as E_{ratio}, f_{side}, or new discrimination variables — benefits automatically from the same corrections without separate tuning.
2. **Correlation preservation**: because the cell energies are corrected before the shower shapes are computed, the correlations between R_η, R_ϕ, and $w_{\eta 2}$ are preserved by construction. Fudge factors, applied independently per variable, can distort these correlations.

With M2 now achieving performance virtually identical to fudge factors, these advantages become decisive: M2 provides the same per-variable quality while additionally preserving the full multivariate structure.

9.5 Wide vs. cell-dependent histogram ranges

Table 10 compares the M2 performance between the wide $[-1, 2]$ and cell-dependent range configurations.

Table 10: Average χ^2/n_{df} for M2 with wide vs. cell-dependent histogram ranges, $Z \to \ell\ell\gamma$ unconverted.

Range configuration	R_η M2	R_ϕ M2	$w_{\eta 2}$ M2
Wide $[-1, 2]$	0.0006	0.0007	0.0005
Cell-dependent	**0.0005**	**0.0006**	0.0005

The cell-dependent configuration provides a consistent improvement, most notably for R_η (0.0005 vs. 0.0006) and R_ϕ (0.0006 vs. 0.0007). The improvement for R_ϕ is the most significant: the tighter peripheral ranges prevent residual noise amplification that affects the 3×3 to 3×7 energy ratio.

For $w_{\eta 2}$, both configurations achieve the same χ^2/n_{df} of 0.0005. This is expected: $w_{\eta 2}$ is computed over a 3×5 window that includes only 15 cells, most of which are central or direct-neighbour cells where the two range configurations produce similar σ estimates.

The wide configuration still produces no catastrophic failures (the GetRMS and removal of the fallback are the primary stabilisation mechanisms), but the cell-dependent ranges provide the additional benefit of optimal resolution and physical regularisation.

For the converted scenario, the difference is more pronounced:

- Wide: R_ϕ M2 = 0.0013, $w_{\eta 2}$ M2 = 0.0013.
- Cell-dependent: R_ϕ M2 = 0.0010, $w_{\eta 2}$ M2 = 0.0010.

The ∼25% improvement for converted photons reflects the more diffuse shower profile, which populates wider tails in the cell-fraction distributions and is more sensitive to range-dependent outlier clipping.

The cell-dependent configuration is recommended as the default.

9.6 Problematic $|\eta|$ regions

Several $|\eta|$ regions exhibit elevated χ^2/n_{df} for all methods:

High $|\eta|$ (> 2.0). The uncorrected MC shows the largest mismodelling for $w_{\eta 2}$ in the forward region: $\chi^2/n_{\mathrm{df}} = 0.027$ at $|\eta| \in [2.0, 2.2)$ and $\chi^2/n_{\mathrm{df}} = 0.030$ at $|\eta| \in [2.2, 2.4)$ for the combined unconverted channel. M2 reduces these to $\chi^2/n_{\mathrm{df}} \approx 0.0003$–$0.0005$, demonstrating that the cell-fraction shift+stretch captures the dominant mismodelling even where the detector geometry differs significantly from the barrel.

Near-crack region ($|\eta| \in [1.30, 1.52)$). The bins immediately adjacent to the calorimeter crack show elevated χ^2/n_{df} for all methods. The $[1.30, 1.37)$ bin has limited data statistics and residual material effects from the crack region that affect the shower profile. All methods produce $\chi^2/n_{\mathrm{df}} \sim 0.01$–$0.02$ here, which is 5–10× worse than the barrel average. The $[1.52, 1.60)$ bin, immediately after the crack, shows a similar pattern. These bins are unlikely to improve without dedicated treatment.

Converted photons. All methods degrade for converted photons relative to unconverted, with the degradation roughly proportional to $\sqrt{N_{\mathrm{unconv}}/N_{\mathrm{conv}}}$, suggesting that statistical precision of the corrections is a limiting factor. For the combined channel, the converted dataset has 45 532 events versus 205 309 unconverted (a factor 4.5 fewer), corresponding to a ∼2× increase in statistical uncertainty on the correction parameters.

9.7 Isolation requirement

Two photon isolation working points are evaluated (see Table 5 in Section 5.3). The loose isolation variant selects ∼205 k data events (unconverted) while the tight variant selects ∼203 k, a reduction of only ∼1%. For converted photons the tight selection yields slightly more events (48 901 vs. 45 532) due to different signal purities of the two working points.

The correction quality in Table 11 shows that the loose isolation variant produces modestly better M2 performance for the unconverted scenario: $\chi^2/n_{\mathrm{df}}(R_\eta) = 0.0005$ vs. 0.0008, $\chi^2/n_{\mathrm{df}}(w_{\eta 2}) = 0.0005$ vs. 0.0008. The R_ϕ variable is comparable between the two (0.0006 in both cases). For the converted and inclusive scenarios the two isolation variants perform similarly.

The slightly lower M2 performance with tight isolation is consistent with the reduced per-cell statistics for the correction derivation: fewer events provide a less precise estimate of the cell fraction distribution, marginally degrading the stretch and shift coefficients. Given the negligible differences in event yield and the similarity in M2 performance, **the loose isolation variant is recommended** as the default, since it maximises the statistics available for correction derivation without sacrificing sample purity relevant for the shower shape study.

9.8 Cross-variant comparison

Table 11 compares the average χ^2/n_{df} across the four pipeline variants (see Section 5.3) for the combined $Z \to \ell\ell\gamma$ channel. Values are averaged over all populated $|\eta|$ bins.

Table 11: Average χ^2/n_{df} comparison across binning and isolation variants. Each cell shows $\langle\chi^2/n_{\mathrm{df}}\rangle$ averaged over all populated $|\eta|$ bins for the $Z \to \ell\ell\gamma$ channel. Lower is better.

		R_η			R_ϕ			$w_{\eta 2}$		
Variant	Scenario	MC	M1	M2	MC	M1	M2	MC	M1	M2
η loose	Unconverted	0.0015	0.0006	0.0005	0.0007	0.0011	0.0006	0.0076	0.0007	0.0005
η loose	Converted	0.0020	0.0014	0.0015	0.0010	0.0010	0.0010	0.0051	0.0011	0.0011
η loose	Inclusive	0.0012	0.0004	0.0004	0.0005	0.0006	0.0004	0.0055	0.0005	0.0004
η tight	Unconverted	0.0015	0.0009	0.0008	0.0007	0.0012	0.0006	0.0076	0.0011	0.0008
η tight	Converted	0.0018	0.0013	0.0013	0.0009	0.0010	0.0009	0.0048	0.0010	0.0010
η tight	Inclusive	0.0012	0.0006	0.0006	0.0004	0.0006	0.0004	0.0054	0.0006	0.0005
$\eta \times p_{\mathrm{T}}$ loose	Unconverted	0.0015	0.0004	0.0004	0.0007	0.0006	0.0004	0.0076	0.0005	0.0005
$\eta \times p_{\mathrm{T}}$ loose	Converted	0.0020	0.0011	0.0013	0.0010	0.0010	0.0009	0.0051	0.0010	0.0010
$\eta \times p_{\mathrm{T}}$ loose	Inclusive	0.0012	0.0003	0.0003	0.0005	0.0004	0.0003	0.0055	0.0003	0.0003
$\eta \times p_{\mathrm{T}}$ tight	Unconverted	0.0015	0.0005	0.0006	0.0007	0.0008	0.0004	0.0076	0.0006	0.0006
$\eta \times p_{\mathrm{T}}$ tight	Converted	0.0018	0.0009	0.0010	0.0009	0.0009	0.0008	0.0048	0.0010	0.0010
$\eta \times p_{\mathrm{T}}$ tight	Inclusive	0.0012	0.0003	0.0004	0.0004	0.0004	0.0003	0.0054	0.0004	0.0004
$\eta \times \langle\mu\rangle$ loose	Unconverted	0.0015	0.0006	0.0005	0.0007	0.0011	0.0006	0.0076	0.0007	0.0005
$\eta \times \langle\mu\rangle$ loose	Converted	0.0020	0.0013	0.0016	0.0010	0.0010	0.0010	0.0051	0.0012	0.0010
$\eta \times \langle\mu\rangle$ loose	Inclusive	0.0012	0.0004	0.0005	0.0005	0.0006	0.0004	0.0055	0.0005	0.0004

Several trends are apparent:

1. **M2 consistently outperforms M1** across all variants and scenarios. This confirms the stabilisation fixes described in Section 9.2 are robust under changes in binning and isolation requirements.

2. **Adding p_{T} bins (η_pt vs. η) improves M2 quality** for unconverted and inclusive photons. The most notable improvement is for R_ϕ: the average χ^2/n_{df} decreases from 0.0006 to 0.0004 (loose) in the unconverted scenario. The 14×6 correction matrices capture pT-dependent mismodelling that a single $|\eta|$-only correction cannot resolve.

3. **For converted photons the improvement from p_{T} binning is smaller.** This is expected: the correction statistics per $(|\eta|, p_{\mathrm{T}})$ bin are thinner for converted photons ($\sim$8 k data events total), and the per-pT correction matrices approach the statistical noise floor.

4. **Loose isolation provides slightly better corrections than tight** for unconverted photons, consistent with the discussion in Section 9.7. For converted and inclusive scenarios the difference is negligible.

5. **The `eta_pt_loose` variant achieves the best overall** χ^2/n_{df} for the inclusive scenario: $\chi^2/n_{\mathrm{df}}(R_\eta) = 0.0003$, $\chi^2/n_{\mathrm{df}}(R_\phi) = 0.0003$, $\chi^2/n_{\mathrm{df}}(w_{\eta 2}) = 0.0003$. This is the recommended configuration for production use.

10 Summary

This note documents a cell-level energy reweighting pipeline for correcting photon shower shape variables (R_η, R_ϕ, $w_{\eta 2}$) in ATLAS Run 3 simulation. Using $Z \to \ell\ell\gamma$ events from 2024 data and MC23e, corrections are derived and applied across four pipeline variants (see Table 4) covering $|\eta|$-only and two-dimensional $(|\eta|, p_{\mathrm{T}})$ correction binning with loose and tight isolation requirements, for three conversion scenarios (unconverted, converted, inclusive).

Two methods are implemented:

- **M1 (flat shift)**: corrects the mean cell energy fraction with exact energy conservation. Reduces the average χ^2/n_{df} for R_η by a factor 2.5 and for $w_{\eta 2}$ by a factor 10.9, but degrades R_ϕ by 1.6× due to the sensitivity of $R_\phi = E(3\times3)/E(3\times7)$ to coherent mean shifts in overlapping windows.

- **M2 (shift+stretch, stabilised)**: additionally matches the cell fraction variance using a linear transformation. Three targeted improvements — histogram-based σ estimation via `GetRMS()`, cell-dependent histogram ranges providing natural regularisation, and removal of the minimum-fraction fallback — eliminated the catastrophic instabilities (χ^2/n_{df} degradations up to 90×) that affected the initial implementation. The stabilised M2 now **outperforms M1 for all three variables**, with improvements of 17% (R_η), 45% (R_ϕ), and 29% ($w_{\eta 2}$).

M2 with cell-dependent histogram ranges is the recommended cell-level correction method. Its average χ^2/n_{df} nearly matches the empirical ATLAS fudge factors: identical for R_η and $w_{\eta 2}$ (0.0005), and within 20% for R_ϕ (0.0006 vs. 0.0005). Unlike fudge factors, M2 preserves inter-variable correlations and automatically improves all observables built from Layer 2 cell energies.

Among the four pipeline variants, the `eta_pt_loose` **variant** — two-dimensional ($|\eta|$, p_{T}) binning with loose isolation — achieves the best correction quality for the combined channel, reaching $\langle\chi^2/n_{\mathrm{df}}\rangle = 0.0003$ for all three shower shape variables in the inclusive scenario. The p_{T} binning provides the largest improvement for R_ϕ (0.0006 → 0.0004 for unconverted loose), where the pT dependence of the cell energy profile is not fully captured by $|\eta|$-only corrections. Loose isolation is preferred over tight, as it maximises the statistics for correction derivation with negligible impact on sample purity.

A cross-source comparison with Francisco Almeida's `photoncellbasedrw` Run 3 output confirms equivalent correction quality. Small residual χ^2/n_{df} differences are fully attributable to different shower shape histogram axis ranges (Francisco uses narrower ranges that exclude the distribution tails, producing systematically lower χ^2/n_{df} values for the same underlying correction).

A Per-Eta Chi-Squared Tables

This appendix contains the per-$|\eta|$-bin χ^2/n_{df} tables for all conversion scenario combinations.

Table 12: Per-$|\eta|$-bin χ^2/n_{df} for $Z \to ee\gamma$, Unconverted scenario. Columns show χ^2/n_{df} between each MC variant and data (cell-computed).

$\lvert\eta\rvert$ bin	R_η				R_ϕ				$w_{\eta 2}$			
(range)	MC	M1	M2	Fud	MC	M1	M2	Fud	MC	M1	M2	Fud
Average	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table 13: Per-$|\eta|$-bin χ^2/n_{df} for $Z \to ee\gamma$, Converted scenario. Columns show χ^2/n_{df} between each MC variant and data (cell-computed).

$\lvert\eta\rvert$ bin	R_η				R_ϕ				$w_{\eta 2}$			
(range)	MC	M1	M2	Fud	MC	M1	M2	Fud	MC	M1	M2	Fud
Average	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table 14: Per-$|\eta|$-bin χ^2/n_{df} for $Z \to ee\gamma$, Conv. & Unconv. scenario. Columns show χ^2/n_{df} between each MC variant and data (cell-computed).

$\|\eta\|$ bin	R_η				R_ϕ				$w_{\eta 2}$			
(range)	MC	M1	M2	Fud	MC	M1	M2	Fud	MC	M1	M2	Fud
Average	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table 15: Per-$|\eta|$-bin χ^2/n_{df} for $Z \to \mu\mu\gamma$, Unconverted scenario. Columns show χ^2/n_{df} between each MC variant and data (cell-computed).

$\|\eta\|$ bin	R_η				R_ϕ				$w_{\eta 2}$			
(range)	MC	M1	M2	Fud	MC	M1	M2	Fud	MC	M1	M2	Fud
Average	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table 16: Per-$|\eta|$-bin χ^2/n_{df} for $Z \to \mu\mu\gamma$, Converted scenario. Columns show χ^2/n_{df} between each MC variant and data (cell-computed).

$\|\eta\|$ bin	R_η				R_ϕ				$w_{\eta 2}$			
(range)	MC	M1	M2	Fud	MC	M1	M2	Fud	MC	M1	M2	Fud
Average	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table 17: Per-$|\eta|$-bin χ^2/n_{df} for $Z \to \mu\mu\gamma$, Conv. & Unconv. scenario. Columns show χ^2/n_{df} between each MC variant and data (cell-computed).

$\|\eta\|$ bin	R_η				R_ϕ				$w_{\eta 2}$			
(range)	MC	M1	M2	Fud	MC	M1	M2	Fud	MC	M1	M2	Fud
Average	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table 18: Per-$|\eta|$-bin χ^2/n_{df} for $Z \to \ell\ell\gamma$, Unconverted scenario. Columns show χ^2/n_{df} between each MC variant and data (cell-computed).

$\|\eta\|$ bin	R_η				R_ϕ				$w_{\eta 2}$			
(range)	MC	M1	M2	Fud	MC	M1	M2	Fud	MC	M1	M2	Fud
[0.00, 0.20)	0.0002	0.0002	0.0001	0.0002	0.0004	0.0003	0.0002	0.0002	0.0008	0.0002	0.0002	0.0003
[0.20, 0.40)	0.0002	0.0001	0.0001	0.0001	0.0003	0.0002	0.0002	0.0002	0.0014	0.0002	0.0002	0.0003
[0.40, 0.60)	0.0003	0.0001	0.0001	0.0002	0.0005	0.0002	0.0003	0.0002	0.0016	0.0002	0.0002	0.0003
[0.60, 0.80)	0.0003	0.0001	0.0001	0.0002	0.0003	0.0003	0.0002	0.0002	0.0016	0.0002	0.0002	0.0003
[0.80, 1.00)	0.0005	0.0001	0.0002	0.0002	0.0007	0.0003	0.0002	0.0003	0.0006	0.0003	0.0002	0.0003
[1.00, 1.20)	0.0004	0.0001	0.0001	0.0002	0.0010	0.0005	0.0003	0.0003	0.0005	0.0004	0.0002	0.0003
[1.20, 1.30)	0.0007	0.0003	0.0003	0.0004	0.0008	0.0006	0.0003	0.0004	0.0010	0.0006	0.0003	0.0004
[1.30, 1.37)	0.0027	0.0018	0.0011	0.0019	0.0014	0.0015	0.0010	0.0014	0.0022	0.0020	0.0021	0.0012
[1.52, 1.60)	0.0014	0.0012	0.0009	0.0008	0.0010	0.0013	0.0010	0.0011	0.0072	0.0010	0.0009	0.0010
[1.60, 1.80)	0.0013	0.0003	0.0003	0.0003	0.0004	0.0006	0.0004	0.0005	0.0080	0.0002	0.0003	0.0003
[1.80, 2.00)	0.0023	0.0003	0.0005	0.0004	0.0005	0.0007	0.0005	0.0004	0.0171	0.0011	0.0006	0.0004
[2.00, 2.20)	0.0034	0.0003	0.0004	0.0006	0.0006	0.0005	0.0003	0.0004	0.0273	0.0004	0.0003	0.0004
[2.20, 2.40)	0.0054	0.0005	0.0006	0.0005	0.0011	0.0010	0.0004	0.0008	0.0296	0.0004	0.0004	0.0005
Average	0.0015	0.0004	0.0004	0.0005	0.0007	0.0006	0.0004	0.0005	0.0076	0.0005	0.0005	0.0005

Table 19: Per-$|\eta|$-bin χ^2/n_{df} for $Z \to \ell\ell\gamma$, Converted scenario. Columns show χ^2/n_{df} between each MC variant and data (cell-computed).

$\|\eta\|$ bin	R_η				R_ϕ				$w_{\eta 2}$			
(range)	MC	M1	M2	Fud	MC	M1	M2	Fud	MC	M1	M2	Fud
[0.00, 0.20)	0.0021	0.0020	0.0031	0.0022	0.0014	0.0018	0.0013	0.0015	0.0021	0.0018	0.0019	0.0018
[0.20, 0.40)	0.0009	0.0009	0.0014	0.0010	0.0007	0.0015	0.0010	0.0011	0.0025	0.0015	0.0013	0.0017
[0.40, 0.60)	0.0011	0.0017	0.0012	0.0016	0.0013	0.0009	0.0013	0.0011	0.0020	0.0013	0.0011	0.0011
[0.60, 0.80)	0.0013	0.0018	0.0020	0.0010	0.0009	0.0010	0.0007	0.0009	0.0017	0.0012	0.0012	0.0010
[0.80, 1.00)	0.0009	0.0007	0.0007	0.0006	0.0006	0.0004	0.0004	0.0005	0.0007	0.0005	0.0005	0.0005
[1.00, 1.20)	0.0006	0.0005	0.0004	0.0004	0.0003	0.0004	0.0004	0.0006	0.0008	0.0011	0.0009	0.0005
[1.20, 1.30)	0.0009	0.0006	0.0007	0.0006	0.0008	0.0006	0.0004	0.0008	0.0011	0.0005	0.0007	0.0006
[1.30, 1.37)	0.0029	0.0021	0.0031	0.0014	0.0023	0.0021	0.0022	0.0016	0.0015	0.0016	0.0016	0.0015
[1.52, 1.60)	0.0017	0.0009	0.0012	0.0011	0.0007	0.0008	0.0009	0.0008	0.0056	0.0009	0.0008	0.0009
[1.60, 1.80)	0.0011	0.0003	0.0003	0.0003	0.0004	0.0006	0.0003	0.0006	0.0052	0.0004	0.0002	0.0004
[1.80, 2.00)	0.0028	0.0006	0.0011	0.0005	0.0005	0.0005	0.0006	0.0007	0.0091	0.0004	0.0004	0.0005
[2.00, 2.20)	0.0042	0.0007	0.0008	0.0006	0.0008	0.0007	0.0005	0.0006	0.0177	0.0009	0.0008	0.0005
[2.20, 2.40)	0.0051	0.0009	0.0010	0.0012	0.0017	0.0014	0.0013	0.0017	0.0164	0.0010	0.0013	0.0009
Average	0.0020	0.0011	0.0013	0.0010	0.0010	0.0010	0.0009	0.0010	0.0051	0.0010	0.0010	0.0009

Table 20: Per-$|\eta|$-bin χ^2/n_{df} for $Z \to \ell\ell\gamma$, Conv. & Unconv. scenario. Columns show χ^2/n_{df} between each MC variant and data (cell-computed).

$\|\eta\|$ bin	R_η				R_ϕ				$w_{\eta 2}$			
(range)	MC	M1	M2	Fud	MC	M1	M2	Fud	MC	M1	M2	Fud
[0.00, 0.20)	0.0002	0.0002	0.0001	0.0002	0.0004	0.0003	0.0001	0.0002	0.0008	0.0002	0.0002	0.0003
[0.20, 0.40)	0.0002	0.0001	0.0001	0.0001	0.0003	0.0002	0.0001	0.0002	0.0013	0.0002	0.0002	0.0003
[0.40, 0.60)	0.0003	0.0001	0.0001	0.0001	0.0005	0.0002	0.0003	0.0002	0.0015	0.0002	0.0002	0.0003
[0.60, 0.80)	0.0003	0.0001	0.0001	0.0002	0.0003	0.0002	0.0002	0.0001	0.0014	0.0002	0.0002	0.0003
[0.80, 1.00)	0.0004	0.0001	0.0001	0.0002	0.0006	0.0003	0.0001	0.0002	0.0005	0.0002	0.0002	0.0003
[1.00, 1.20)	0.0003	0.0001	0.0001	0.0002	0.0006	0.0003	0.0002	0.0002	0.0004	0.0003	0.0002	0.0003
[1.20, 1.30)	0.0005	0.0002	0.0002	0.0002	0.0005	0.0004	0.0002	0.0002	0.0008	0.0003	0.0003	0.0004
[1.30, 1.37)	0.0018	0.0010	0.0007	0.0010	0.0009	0.0010	0.0008	0.0008	0.0014	0.0008	0.0007	0.0007
[1.52, 1.60)	0.0011	0.0007	0.0006	0.0006	0.0004	0.0005	0.0005	0.0004	0.0053	0.0006	0.0005	0.0006
[1.60, 1.80)	0.0010	0.0001	0.0002	0.0001	0.0002	0.0002	0.0002	0.0002	0.0054	0.0001	0.0001	0.0002
[1.80, 2.00)	0.0022	0.0002	0.0004	0.0003	0.0003	0.0003	0.0004	0.0002	0.0115	0.0005	0.0004	0.0002
[2.00, 2.20)	0.0030	0.0002	0.0003	0.0002	0.0004	0.0004	0.0003	0.0003	0.0202	0.0003	0.0003	0.0003
[2.20, 2.40)	0.0047	0.0005	0.0004	0.0005	0.0009	0.0006	0.0006	0.0007	0.0211	0.0005	0.0004	0.0004
Average	0.0012	0.0003	0.0003	0.0003	0.0005	0.0004	0.0003	0.0003	0.0055	0.0003	0.0003	0.0003

B Cross-Source Comparison Per-Eta Tables

This appendix contains the per-$|\eta|$-bin χ^2/n_{df} comparison between Francisco's `photoncellbasedrw` Run 3 output, the wide $[-1, 2]$ range configuration, and the cell-dependent range configuration, for each shower shape variable individually.

Table 21: χ^2/ndf comparison for R_η (unconverted, llgamma). Francisco has a single cell-based correction (shown as Corr.); M1/M2 are defined for Wide and Cell-dep. only.

	Francisco / run3			Wide $[-1, 2]$				Cell-dep.			
$\lvert\eta\rvert$	MC	Corr.	Fudge	MC	M1	M2	Fudge	MC	M1	M2	Fudge
$[0.00, 0.20)$	0.0002	0.0002	0.0003	0.0002	0.0002	0.0002	0.0003	0.0002	0.0002	0.0002	0.0003
$[0.20, 0.40)$	0.0002	0.0002	0.0001	0.0002	0.0001	0.0001	0.0001	0.0002	0.0001	0.0001	0.0001
$[0.40, 0.60)$	0.0002	0.0002	0.0002	0.0003	0.0002	0.0002	0.0002	0.0003	0.0002	0.0002	0.0002
$[0.60, 0.80)$	0.0003	0.0002	0.0002	0.0003	0.0002	0.0002	0.0002	0.0003	0.0002	0.0002	0.0002
$[0.80, 1.00)$	0.0004	0.0002	0.0002	0.0005	0.0002	0.0005	0.0002	0.0005	0.0002	0.0002	0.0002
$[1.00, 1.20)$	0.0004	0.0002	0.0002	0.0004	0.0001	0.0001	0.0002	0.0004	0.0001	0.0002	0.0002
$[1.20, 1.30)$	0.0005	0.0004	0.0004	0.0007	0.0004	0.0013	0.0004	0.0007	0.0004	0.0005	0.0004
$[1.30, 1.37)$	0.0023	0.0016	0.0013	0.0027	0.0011	0.0014	0.0015	0.0027	0.0011	0.0017	0.0015
$[1.52, 1.60)$	0.0013	0.0016	0.0009	0.0014	0.0027	0.0015	0.0008	0.0014	0.0027	0.0015	0.0008
$[1.60, 1.80)$	0.0012	0.0005	0.0003	0.0013	0.0005	0.0004	0.0003	0.0013	0.0005	0.0004	0.0003
$[1.80, 2.00)$	0.0019	0.0006	0.0004	0.0023	0.0005	0.0008	0.0004	0.0023	0.0005	0.0009	0.0004
$[2.00, 2.20)$	0.0030	0.0004	0.0005	0.0034	0.0004	0.0005	0.0006	0.0034	0.0004	0.0004	0.0006
$[2.20, 2.40)$	0.0049	0.0005	0.0005	0.0054	0.0006	0.0007	0.0005	0.0054	0.0006	0.0007	0.0005
Average	**0.0013**	**0.0005**	**0.0004**	**0.0015**	**0.0006**	**0.0006**	**0.0004**	**0.0015**	**0.0006**	**0.0005**	**0.0004**

Table 22: χ^2/ndf comparison for R_ϕ (unconverted, llgamma). Francisco has a single cell-based correction (shown as Corr.); M1/M2 are defined for Wide and Cell-dep. only.

	Francisco / run3			Wide $[-1, 2]$				Cell-dep.			
$\lvert\eta\rvert$	MC	Corr.	Fudge	MC	M1	M2	Fudge	MC	M1	M2	Fudge
$[0.00, 0.20)$	0.0004	0.0003	0.0002	0.0004	0.0005	0.0003	0.0002	0.0004	0.0005	0.0003	0.0002
$[0.20, 0.40)$	0.0003	0.0003	0.0002	0.0003	0.0004	0.0003	0.0002	0.0003	0.0004	0.0003	0.0002
$[0.40, 0.60)$	0.0004	0.0003	0.0002	0.0005	0.0005	0.0004	0.0002	0.0005	0.0005	0.0003	0.0002
$[0.60, 0.80)$	0.0003	0.0002	0.0002	0.0003	0.0004	0.0003	0.0002	0.0003	0.0004	0.0003	0.0002
$[0.80, 1.00)$	0.0006	0.0002	0.0002	0.0007	0.0005	0.0003	0.0003	0.0007	0.0005	0.0003	0.0003
$[1.00, 1.20)$	0.0009	0.0004	0.0002	0.0010	0.0007	0.0004	0.0003	0.0010	0.0007	0.0004	0.0003
$[1.20, 1.30)$	0.0007	0.0004	0.0004	0.0008	0.0010	0.0013	0.0004	0.0008	0.0010	0.0004	0.0004
$[1.30, 1.37)$	0.0014	0.0016	0.0014	0.0014	0.0016	0.0013	0.0015	0.0014	0.0016	0.0011	0.0015
$[1.52, 1.60)$	0.0008	0.0009	0.0008	0.0010	0.0036	0.0012	0.0011	0.0010	0.0036	0.0011	0.0011
$[1.60, 1.80)$	0.0003	0.0005	0.0004	0.0004	0.0011	0.0007	0.0005	0.0004	0.0011	0.0006	0.0005
$[1.80, 2.00)$	0.0004	0.0006	0.0003	0.0005	0.0011	0.0011	0.0004	0.0005	0.0011	0.0009	0.0004
$[2.00, 2.20)$	0.0004	0.0006	0.0003	0.0006	0.0009	0.0006	0.0004	0.0006	0.0009	0.0005	0.0004
$[2.20, 2.40)$	0.0008	0.0008	0.0006	0.0011	0.0014	0.0007	0.0008	0.0011	0.0014	0.0007	0.0008
Average	**0.0006**	**0.0005**	**0.0004**	**0.0007**	**0.0011**	**0.0007**	**0.0005**	**0.0007**	**0.0011**	**0.0006**	**0.0005**

References

[1] F. Almeida, "Shower shape corrections using cell-level reweighting", ATL-COM-PHYS-2021-640 (2021).

[2] M. Fernandes da Silva, "Photon shower shape studies with Run 3 data", ATL-COM-PHYS-2025-662 (2025).

Table 23: χ^2/ndf comparison for $w_{\eta 2}$ (unconverted, llgamma). Francisco has a single cell-based correction (shown as Corr.); M1/M2 are defined for Wide and Cell-dep. only.

	Francisco / run3			Wide $[-1, 2]$				Cell-dep.			
$\lvert\eta\rvert$	MC	Corr.	Fudge	MC	M1	M2	Fudge	MC	M1	M2	Fudge
$[0.00, 0.20)$	0.0008	0.0002	0.0004	0.0008	0.0003	0.0002	0.0005	0.0008	0.0003	0.0002	0.0005
$[0.20, 0.40)$	0.0013	0.0003	0.0003	0.0014	0.0004	0.0003	0.0003	0.0014	0.0004	0.0003	0.0003
$[0.40, 0.60)$	0.0015	0.0002	0.0003	0.0016	0.0003	0.0002	0.0003	0.0016	0.0003	0.0003	0.0003
$[0.60, 0.80)$	0.0015	0.0002	0.0003	0.0016	0.0004	0.0003	0.0003	0.0016	0.0004	0.0003	0.0003
$[0.80, 1.00)$	0.0005	0.0002	0.0003	0.0006	0.0004	0.0003	0.0003	0.0006	0.0004	0.0002	0.0003
$[1.00, 1.20)$	0.0005	0.0003	0.0003	0.0005	0.0006	0.0003	0.0003	0.0005	0.0006	0.0003	0.0003
$[1.20, 1.30)$	0.0009	0.0004	0.0004	0.0010	0.0007	0.0003	0.0004	0.0010	0.0007	0.0004	0.0004
$[1.30, 1.37)$	0.0024	0.0016	0.0020	0.0022	0.0020	0.0015	0.0016	0.0022	0.0020	0.0019	0.0016
$[1.52, 1.60)$	0.0066	0.0009	0.0010	0.0072	0.0014	0.0011	0.0010	0.0072	0.0014	0.0011	0.0010
$[1.60, 1.80)$	0.0072	0.0004	0.0003	0.0080	0.0004	0.0004	0.0003	0.0080	0.0004	0.0004	0.0003
$[1.80, 2.00)$	0.0134	0.0006	0.0004	0.0171	0.0015	0.0008	0.0004	0.0171	0.0015	0.0008	0.0004
$[2.00, 2.20)$	0.0230	0.0003	0.0003	0.0273	0.0005	0.0004	0.0004	0.0273	0.0005	0.0003	0.0004
$[2.20, 2.40)$	0.0300	0.0005	0.0004	0.0296	0.0005	0.0005	0.0005	0.0296	0.0005	0.0005	0.0005
Average	**0.0069**	**0.0005**	**0.0005**	**0.0076**	**0.0007**	**0.0005**	**0.0005**	**0.0076**	**0.0007**	**0.0005**	**0.0005**